

E-Commerce D2C Sales OS

Exhaustive Feature-by-Feature Reference: Frontend Views & Backend Architecture

PART 1: BACKEND FEATURES & INTERNAL ENGINES

Feature B-01: Multi-Tenant Subdomain Authentication & Security (routes/auth.js)

- **Tenant Company Registration (POST /api/auth/register):** Accepts name, subdomain, ownerEmail, and password. The system sanitizes the subdomain into lowercase alphanumeric characters, verifies no collisions exist in MongoDB, generates a cryptographic salt (10 rounds via bcryptjs), and hashes the password. Once saved, it signs an asynchronous JWT token containing { companyId, subdomain, email } with a 7-day expiry.
- **Subdomain-Isolated Login (POST /api/auth/login):** Validates tenant credentials by scoping queries to both subdomain and email. Verifies bcrypt hashes and returns the signed JWT token. This ensures total zero-trust tenant isolation across stores.

Feature B-02: Product Catalog & Tax Computation Engine (routes/products.js)

- **Multi-File Multer Upload (POST /api/products/create):** Enforces authMiddleware (extracts and verifies Bearer token). Configured with disk storage writing to /uploads/ with unique timestamp hashing. Accepts up to 5 image uploads simultaneously.
- **Automated GST Profile Mapping:** Stores base price, stock count, and GST bracket (0%, 5%, 12%, 18%, 28%). Computes dynamic retail value: $\text{retailPrice} = \text{price} * (1 + \text{gstPercent} / 100)$.
- **Tenant Catalog Query (GET /api/products/list):** Accepts query filters companyId and status. Returns products sorted by createdAt: -1 (newest arrivals first).

Feature B-03: Algorithmic Order Management & RTO Risk Engine (routes/orders.js)

- **Automated Tax & Subtotal Breakdown:** When an order is placed via POST /api/orders/create, the backend iterates over line items, retrieves each product's base price and GST percentage, computes total tax and gross payable, and writes normalized sub-records.
- **Heuristic Fraud Evaluation Algorithm (calculateRtoScore):** Evaluates fraud probability on a 0-100 scale:
 1. *Baseline Confidence (+15):* Initial score assigned to all transactions.
 2. *Payment Mode Weighting:* COD (+45) due to high refusal rates in Indian e-commerce vs UPI/Razorpay (+2 to +5).
 3. *High-Value COD Threshold:* If payment is COD and cart value exceeds Rs. 2,000, adds +25 points.
 4. *Repetitive Digit Pattern:* Regular expression test `/^(\d){1,9}$` checks for fake numbers (e.g. 9111111111), adding +30 points.
 5. *Automated Status Quarantining:* Orders with RTO score > 75 are automatically assigned status = 'rto' for merchant review.
- **Analytics & Stats Engine (GET /api/orders/stats):** Aggregates counts for total orders, RTO flagged orders, delivered shipments, and active fulfillments.

Feature B-04: Automated Shiprocket Logistics Dispatch (routes/orders.js)

- **Shipment Manifesting (POST /api/orders/shiprocket/ship):** Accepts orderId. Integrates with Shiprocket Logistics services to generate a unique Air Waybill (AWB) tracking number (e.g. SR839201948) and transitions order status from valid to shipped.

Feature B-05: Abandoned Cart Recovery & Cron Daemon (cron.js & routes/orders.js)

- **Cart Drop-off Logger (POST /api/orders/abandoned/check):** Records uncompleted checkout sessions with customer phone, target product reference, and cart value with recovery schedule reminderDays: [0, 1, 3].
- **Daily Background Cron Daemon (node-cron):** Scheduled at 0 8,15,20 * * * (daily at 8:00 AM, 3:00 PM, and 8:00 PM).
 1. Fetches all unrecovered carts where recovered === false.
 2. Calculates elapsed time in days: $\text{diffDays} = \text{Math.floor}((\text{now} - \text{createdAt}) / (24 * 60 * 60 * 1000))$.
 3. If reminderDays.includes(diffDays), automatically triggers recovery dispatches across WhatsApp and Fast2SMS.

Feature B-06: Multilingual WhatsApp & UPI Deep Linking Engine (routes/whatsapp.js)

- **Meta WhatsApp Cloud API Integration (POST /api/whatsapp/abandoned):** Dispatches WhatsApp messages to customers with instant 1-tap UPI payment deep links (`upi://pay?pa=merchant@upi&pn;=D2CStore&am;={amount}&cu;=INR&tn;=CompleteCart_{id}`).
- **Dynamic Localization Engine:** Interpolates product title and payable amount into three dialect profiles: English, Marathi (Regional), and Hindi (National).
- **Automated WhatsApp Invoice Generator (POST /api/whatsapp/invoice):** Sends order confirmation receipts and tracking summaries to customer phones.

Feature B-07: Tally ERP Bi-Directional Inventory Synchronization (routes/tally.js)

- **Bulk Stock Synchronizer (POST /api/tally/sync):** Accepts inventory data arrays mapping SKUs or tallyItemId to live warehouse stock numbers and prices. Atomically updates the Product catalog.
- **Audit Trail Logger (GET /api/tally/logs):** Records each synchronization event in the TallySync collection with timestamps, item counts, and status.

Feature B-08: Razorpay HMAC-SHA256 Payment Verification (routes/payment.js)

- **Cryptographic Payment Reconciliation (POST /api/payment/verify):** Computes HMAC-SHA256 signature over `razorpay_order_id + '|' + razorpay_payment_id` using RAZORPAY_SECRET. Verifies signature authenticity and updates order status to paid.

Feature B-09: Financial Audit & Year 2026 Excel Export Generator (routes/audit.js)

- **Native Excel Streaming (GET /api/audit/export?year=2026):** Uses exceljs to construct and stream a styled .xlsx workbook with custom headers and borders, detailing Title, SKU, Total, GST, and Status for compliance and tax filings.

PART 2: FRONTEND NEXT.JS VIEWS & USER EXPERIENCE

Feature F-01: Multi-Tenant Login & Brand Onboarding (/login)

- **Interactive Mode Toggle:** Switch between 'Sign In' and 'Register New Tenant Store' with real-time field transitions.
- **Subdomain Validator:** Automatically formats brand subdomains into valid domain handles with live .d2cos.com suffix indicator.
- **Session Hydration:** Saves JWT token, companyId, and brand name into browser storage and seamlessly redirects to the Dashboard.

Feature F-02: Analytics Dashboard & RTO Fraud Control Screen (/dashboard)

- **Live Metric Cards:** Displays Total Products in catalog, Total Orders with delivered breakdown, Abandoned Carts in recovery, and RTO Risk Percentage.
- **RTO High-Risk Quarantine Table:** Directly queries `/orders/list?status=rto` to display suspicious orders with color-coded risk bars (0-100) and customer contact numbers.
- **One-Click 2026 Audit Export Button:** Direct trigger that invokes the backend Excel generator and downloads Audit_Export_2026.xlsx.

Feature F-03: Catalog & Multi-Image Product Management (/products)

- **Product Upload Form:** Form inputs for Title, unique SKU, Base Price (Rs.), GST selection dropdown (0%, 5%, 12%, 18%, 28%), and Stock Quantity.
- **Multer Multi-Image Uploader:** Supports selecting up to 5 image files simultaneously with preview counts.
- **Live Catalog Gallery Table:** Displays active products with square image thumbnails, SKU tags, base price, tax rate, computed retail price with GST, and low-stock indicators.

Feature F-04: Order Operations & Shiprocket Fulfillment View (/orders)

- **Comprehensive Order Table:** Displays Order ID, creation date, customer name, phone number, line items, total with GST breakdown, and payment mode.
- **One-Click 'Mark Shipped' Action:** Directly triggers Shiprocket AWB generation and displays the live tracking number on screen.
- **One-Click 'Send Invoice WhatsApp' Action:** Dispatches instant order receipts to customer WhatsApp numbers.
- **Interactive Order Simulator Modal:** Allows operators to test orders with custom names, phones, products, and payment modes (COD vs UPI) to verify RTO scoring in real time.

Feature F-05: Abandoned Cart Recovery Control Center (/abandoned)

- **Dropped Cart Monitor:** Shows customers who abandoned checkout, the items they left behind, and the total cart value.
- **Language Selector Bar:** Allows store operators to preview and dispatch recovery messages in English, Marathi, or Hindi.
- **Instant 'Send WhatsApp Now' Trigger:** Manually fires the WhatsApp recovery API with live message previews and 1-tap UPI deep links.
- **Cart Drop-off Simulator:** Lets operators simulate cart abandonments with custom phone numbers and amounts.

Feature F-06: Tally ERP Warehouse Synchronization Screen (/tally)

- **ERP Connector Status Card:** Displays real-time connection health, last sync timestamp, and total items updated.
- **'Trigger Tally Sync Now' Button:** Initiates an immediate ledger synchronization with physical warehouse stock.
- **Historical Audit Log Table:** Displays past sync records with sync ID, timestamp, updated item count, and success status.

Feature F-07: Global System & Integration Settings (/settings)

- **Automated Gateway Toggles:** Interactive switches to enable/disable WhatsApp Recovery, SMS Notifications (Fast2SMS), and UPI AutoPay Mandates.
- **Store Language Preference Selector:** Select default store language between English, Marathi, and Hindi.
- **Gateway Credentials Form:** Manage Razorpay Key ID, Shiprocket Email, and Shiprocket Password.
- **'Test Alert' Simulator:** Triggers a simulated notification cycle across active channels.

Feature B-10 & F-08: Complete Cross-Reference Feature Matrix

Feature Component	Layer	Endpoint / Path	Key Capability
Tenant Authentication	Backend	POST /api/auth/register, /login	Bcrypt hash, subdomain isolation, signed JWT tokens
Product Tax Management	Full-Stack	/products & /api/products/create	Multer 5-file upload, 0-28% GST calc, catalog gallery
RTO Fraud Scoring	Backend	POST /api/orders/create	Algorithmic COD, cart value, and phone pattern scoring
Logistics Fulfillment	Full-Stack	/orders & /api/orders/shiprocket/ship	Shiprocket AWB tracking generation & status updates
Abandoned Recovery	Full-Stack	/abandoned & /api/whatsapp/abandoned	0,1,3 day cron, 1-tap UPI deep links, WhatsApp Cloud API
Multilingual Messages	Full-Stack	/settings & routes/whatsapp.js	English, Marathi, and Hindi translation templates
Tally ERP Stock Sync	Full-Stack	/tally & /api/tally/sync	Bi-directional stock ledger sync and TallySync audit log
Year 2026 Audit Export	Full-Stack	/dashboard & /api/audit/export	Native ExcelJS .xlsx export with Title, SKU, Total, GST
Payment Reconciliation	Backend	POST /api/payment/verify	Razorpay HMAC-SHA256 signature verification to paid